Investigating Transient Chaos and Immediate Geometric Transitions in the Classification of Non-Linear Equations

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Abstract

This paper introduces a structural classification coefficient $\xi = ((1 - |sgn(D)|)(\frac{1}{K+1})) + (|sgn(D)| \times (1 + \frac{1}{2(D+K)}))$. Using a custom digital bifurcation diagram simulator, the study tracks and analyses the chaotic behaviours of multiple non-linear equations over continuous ranges of parameters. The simulator ran 1000 settling iterations followed by 500 recorded steps per value. For the logistic map rx(1-x) and rsin(x), smooth unimodal maps were classified under $\xi = 0.5$. It demonstrated continuous pitchfork branching routes to chaos that scale precisely according to the universal Feigenbaum constant of $\delta = 4.669$. Smooth multimodal maps classified under $0 < \xi < 0.5$, for cubic and quintic equations like $r(x^5-x^3+x)$, showcase overlapping wave-like geometric structures and sudden vertical attractor expansions triggered by interior crises. Piecewise linear maps were categorised under $1.0 < \xi < 1.5$, for the linear tent map r(1-2|x-0.5|) and the absolute value map 1-r|x|, bypass the period-doubling cascade entirely, transitioning directly from a single stable fixed point into a solid chaotic wedge characterised by uniform point density and distinct unvisited lens-shaped regions. These quantitative findings establish that the classification coefficient functions as an accurate baseline for classifying non-linear chaotic dynamics based on the visual properties of the bifurcation diagrams.

Summary

This research paper investigates unpredictable systems by analyzing bifurcation diagrams, which are visual tracking charts showing how a simple mathematical equation shifts from a steady, repeating state into complete randomness. For example, tracking a rabbit population in an isolated system shows this progression. If the reproduction rate sits at a low level, the population stabilizes at a single fixed number year after year. Increasing this growth parameter causes the population to fluctuate, alternating between two distinct values every other year, then four values, and eventually shifting into complete randomness where the population number becomes impossible to predict.To build this diagram, a computer program runs a population formula thousands of times using a specific growth number, discards the initial unstable fluctuations, and plots the final stable points as dots on a graph. The computer then slightly increases the growth control number and repeats the process, creating a continuous visual map of dots showing how the system forks and shifts.Two main structural shapes inside these mathematical formulas directly influence the final visual patterns of the tracking charts. The first property involves the presence of sharp corners, labeled as D, representing sudden direction changes in the equation. The second property counts the number of smooth peaks and valleys, labeled as K, which reflect the rounded high points and low points of the curve. This study introduces a single tracking number called a classification coefficient, xi, combining these geometric shapes to group unpredictable equations into three distinct structural classes. Class 1 equations represent smooth curves with a single peak, yielding a tracking number of exactly 0.5. These systems approach chaos through a predictable, step-by-step splitting process where a single stable line forks into two lines, then four lines, and continues splitting at a universal geometric rate until complete chaos fills the graph. Class 2 equations encompass smooth curves with multiple peaks and valleys, resulting in tracking numbers between 0 and 0.5. These systems follow a similar line-splitting pattern to the first class, but they experience sudden, wide expansions where the random zones instantly widen to create wave-like density patterns across the chart. Class 3 equations consist of sharp, connected straight lines with tracking numbers sitting between 1.0 and 1.5. Since these formulas contain sharp corners instead of smooth curves, they skip the gradual line-splitting process entirely, forcing the system to jump instantly from a single steady population value straight into a solid wedge of total chaos.

1 Introduction

In quantitative finance, predicting accurate stock price patterns is one of the most challenging problems. Financial markets are shaped by a complex interplay of macroeconomic conditions, investor sentiment, and information asymmetry. This contributes to the price behaviour of a stock that is highly nonlinear, non-stationary, and sensitive to unobserved variables. Despite this complexity, even marginal improvements in forecasting accuracy can translate into meaningful reductions in investment risk and more efficient capital allocation. This motivated a research interest in mathematical models within the framework of dynamical systems to capture the underlying structure of price dynamics. A dynamical system is a mathematical framework in which the state changes over time according to a set of mathematical rules [Shirali, 2002]. These systems can be modelled as either discrete, in which the state transitions at fixed intervals, or continuous, in which the state variables evolve smoothly over time. This is often written as $x_{n+1}=f(x_n)$. In a continuous dynamical system, change happens smoothly over time and is usually described using differential equations $\frac{dx}{dt} = f(x)$ [Shirali, 2002] . These types of systems are based on the assumption that if we know the initial state of a system exactly, then the system should be able to predict its future behaviour [Natarajan, 2000a]. However, in nonlinear dynamic systems, a simple deterministic equation updates a system step by step, with the output of one step becoming the input for the next. [Shirali, 2002]

In mathematical iteration, the initial input is designated as the seed value. When a function is iteratively applied to a seed value, it generates a chronological sequence of values called the orbit of the seeded value, which allows one to understand the behaviour of the function when it is being iterated. These orbits can behave differently depending on the nature of the mapping function. If an orbit value returns exactly to its starting point after a finite number of steps, it forms a closed loop termed a cycle. These are designated as n-cycles, meaning the system state repeats identically with a fixed period of n iterations [Shirali, 2002]. In the long-term behaviour of an orbit, fixed points in dynamical systems often describe points where the system may settle after many iterations. In one-dimensional maps, an attracting fixed point is when the iteration settles when the magnitude of the derivative at the fixed point satisfies $|f'(x\ ^*)| < 1$ whereas in a repelling fixed point, the values move away when $|f'(x\ ^*)| > 1$ [Shirali, 2012]. Prior research had assumed that only unimodal mappings generate universal bifurcation diagrams, assuming that a single maximum value is necessary for period-doubling to produce a bifurcation diagram.

In the case of unimodal maps, which have a single peak, this repeated process can cause major changes in the behaviour of the system. As the parameter changes, the system may move from a stable fixed point to a series of period-doubling changes, and eventually into chaos [Feigenbaum, 1979]. It can produce highly complex and chaotic behaviour, making prediction difficult [Natarajan, 2000a]. The chaotic transition of the nonlinear system $x_{n+1} = rx_n(1-x_n)$ is traditionally characterised through a logistic map and its bifurcation diagram. For example, the system initially described by $y = x$ is altered to $y = x^2$, which means initially, the system has one solution for every position. Later, it changes to two solutions, corresponding to period-doubling behaviour, and the system bifurcates [Natarajan, 2000b]. The generated model predicts that further doubling would occur at each of the two values, generating four, then eight, sixteen, and thirty-two values. The bifurcations become so dense and are eventually indistinguishable from chaos.

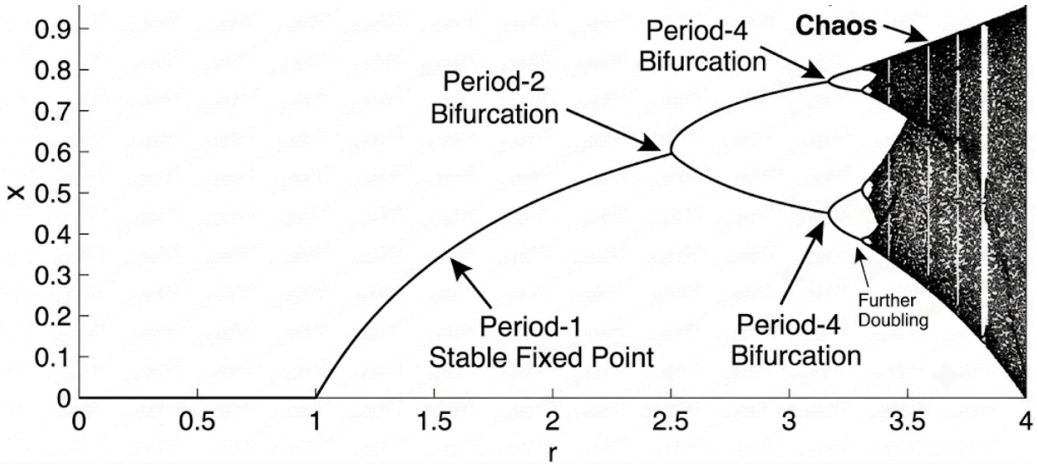


Figure 1: Annotated bifurcation diagram of rx(1-x) between 0 and 4 to show period doubling and chaotic properties in bifurcation diagrams [Vanderbilt University, 2020]

In Figure 1, the period-doubling cascade serves as the geometric pathway toward turbulence, demonstrating the universal constants of Feigenbaum, which apply to all chaotic systems and depend on only one driving parameter. This constant states that period doubling occurs at regular intervals, with the limit of the ratio of the widths of consecutive parameter windows approaching $\sigma = 4.669$ [Feigenbaum, 1979]. As the control parameter advances beyond these shrinking intervals, the infinite splitting of

periodic orbits finishes the cascade and forces the system to cross a critical threshold, where it eventually reaches complete deterministic chaos.

If a_n denotes the parameter value at which a period-2^n orbit first appears (the nth period-doubling bifurcation).

$$\delta = \lim_{n\to\infty} \frac{a_n - a_{n-1}}{a_{n+1} - a_n} \approx 4.669201609 \ldots$$

Feigenbaum found that the parameter intervals obey and that spatial (phase-space) scaling near the critical point is governed by

$$\alpha = \lim_{n\to\infty} \frac{d_n}{d_{n+1}} \approx -2.502907875 \ldots$$

where d_n measures the half-width (or separation) of the stable period-2^n orbit about the critical point. These constants δ and α are universal for the class of maps with the same order of stationary points, so families as different as the logistic map $f_a(x)=ax(1-x)$ and $f_a(x)=a\sin(x)$ share the same limiting ratios even though their individual a_n value differ.[Feigenbaum, 1979]

The work done by Robert M. May, the equation $x_{n+1} = rx_n(1-x_n)$ represents the discrete logistic map, which is the established model for studying chaotic dynamics in the quadratic family [May, 1976]. In this system, the state variable is the normalised unit interval, and the growth rate serves as the control parameter, for example, if the mapping function forms an inverted parabola in phase space, reaching its maximum exactly at the stationary point x = 0.5. Since the maximum value evaluates to $g(0.5) = \frac{r}{4}$, which strictly bounds the iterations within the normalised unit interval, the growth rate acts as the control parameter that regulates the maximum value of the function [Shirali, 2002]. Under this condition, the function maps the interval back onto itself, ensuring that all subsequent iterations remain trapped within a fixed geometric boundary and hence converge to a point. The structural properties and equilibrium states of this unimodal family are completely dictated by the quantitative value of the parameter (r). Resolving

the characteristic fixed-point equation requires calculating the intersection where the parabola crosses the line y=x. This calculation reveals that the system possesses two distinct real algebraic fixed points, namely: $x_1 = 0$ and $x_2 = \frac{r-1}{r}$. The shifting stability of these points produced predictable trends as the parameter is adjusted across specific mathematical regimes [Shirali, 2002].

To quantify the chaos in a system, the Lyapunov exponent was used to measure the rate of separation of two nearby phase points as they follow the trajectory determined by their respective initial conditions. A Lyapunov exponent measures a system's predictability and sensitivity to changes in its initial conditions [Sprott, 2003]. The exponent is the average logarithmic rate of separation or convergence of two nearby points of two time series X_t and Y_t separated by an initial distance.

$$\lim_{x \to \infty} \frac{1}{n} \sum_{i=1}^{n} ln \left| \frac{\Delta R_i}{\Delta R_o} \right|$$

In a chaotic system, the Lyapunov exponent λ must be positive, quantifying the sensitivity to initial conditions. A positive Lyapunov exponent means that two nearby points will separate exponentially over time, indicating the system's chaotic behaviour. It is found that in chaotic regions, trajectories are organised by a strange attractor and are statistically described by an invariant density rather than a single periodic orbit. For certain parameter values, this density follows the U-shaped Arc Sine law [Border, 1991]. The chaotic regimes are also interrupted by periodic windows, during which stable cycles, such as period-three orbits, emerge via tangent bifurcations. These windows are self-similar and contain smaller replicas of the period-doubling cascade [Keener, 1980 ,Blackmore, 1986].

While prior research has primarily focused on a single chaotic map or an isolated type of bifurcation analysis involving unimodal functions, this investigation explores system dynamics by systematically mapping, analyzing, and contrasting the properties of chaos across different nonlinear iterative equations. By examining how structural variations influence bifurcation boundaries, tracking universal geometric scaling, and evaluating statistical distributions within chaotic regions, this paper constructs an analytical framework designed to identify properties of bifurcation diagrams and establish a predictive baseline for nonlinear systems. In this paper, we developed a classification coefficient that can be used to classify equations into different classes based on its value. Each class of equations displays unique features within its bifurcation diagrams while also displaying consistent traits across the class as a whole, alongside distinct variations between the different classes. This coefficient successfully quantifies the visual properties of the logistic map.

2 Methods

In this investigation, a bifurcation diagram simulator was developed for real-time simulations and comparative analysis of discrete-time one-dimensional dynamical systems. The experimental process is initiated by inputting a specific mapping function, such as the Logistic, Sine, and Quadratic Map, into the simulation. With the given starting value, the simulation ran a two-stage loop to determine the settled value. The simulator then systematically varies a primary control parameter, such as the growth rate or tuning constant (r), over a continuous range along the x-axis, while plotting the converged iteration values along the y-axis to capture the long-term evolution of the bifurcation diagram. The simulator processes an initial sequence of 1,000 iterations without recording any data to avoid influencing long-term data. This settling phase was considered to discard short-term behaviours that could force the orbit trajectory onto its permanent chaotic attractor. Afterwards, the simulator ran 500 iterations, and the precise coordinates were recorded to identify the patterns and trends of a specific used function.

The simulation engine mapped the data across three synchronised analytical tools to reveal the underlying properties of the system. The first tool was an interactive cobweb visualizer that plotted the equation directly against the diagonal line y = x. When an orbit is launched from a designated starting point, the visualizer traces its path step by step as it oscillates between the function's curve and the diagonal line. This layout revealed the exact geometric mechanism behind the system's behaviour. The cobweb plot makes it easy to see whether the orbit is being pulled into a single stable fixed point or locked into a repeating periodic cycle,or moving into a region of chaos.

The same data is also fed into a bifurcation diagram generator, a visual tool that shows how the long-term behaviour of a dynamical system changes with a single parameter. The long-term recorded positions of the final converged iteration values were plotted by this diagram on the y-axis against the gradually shifting control parameter on the x-axis. In the simulation the tracking points were plotted with high transparency to create darker, concentrated bands where the dots overlap frequently. This shading technique highlights the exact zones where the chaotic system spends most of its time, revealing shapes and dense behavioral trends clearly. The exact boundaries where stability breaks down are specifically designed to be uncovered by this global view, which displays the precise split points of period-doubling cascades and the sudden appearance of clean, white gaps within the chaos. These gaps represent stable periodic windows, such as the prominent period-three window. The simulation also features interactive controls, including zoom in, zoom out, full screen, a coordinate tracker, and click-and-drag navigation. Finally, the system's behaviour is mapped by the simulation onto a time series and a Lyapunov spectrum monitor. This tool tracks the system's back-to-back positions over time and measures the exponential

separation of two trajectories that start incredibly close to one another. This provides numerical proof to quantify and identify regions of chaos.

Before inputting the equations into the bifurcation diagram simulator, we first had to classify them into different categories. Therefore, a classification coefficient, denoted as ξ, was developed to group these equations into distinct categories based on structural features. This coefficient translates features of a map function, specifically its differentiability and stationary points, to classify equations based on the shape of their bifurcation diagrams. The coefficient relies on two integers extracted directly from the map equation, $x_{n+1} = f(x_n)$, across a specific domain. The first parameter is the sharpness metric denoted as D, which quantifies the number of points where the first derivative f'(x) is discontinuous, undefined, or exhibits a finite jump discontinuity. For smooth analytical functions such as polynomials or classical trigonometric functions, D equals zero. For piecewise functions, or those incorporating absolute values, modulo operators, or floor functions, D is greater than or equal to one, pinpointing the exact number of sharp kinks or structural ruptures. The second parameter K measures the number of stationary points. The value of K can be calculated by counting the points where the first derivative equals zero and the second derivative does not equal zero. Using the values of D and K, the coefficient can be defined as

$$\xi = ((1 - |sgn(D)|)(\frac{1}{K+1})) + (|sgn(D)| \times (1 + \frac{1}{2(D+K)}))$$

The classifying coefficient divides all equations into certain classes:

- Class 1 are the functions with smooth unimodal maps. An equation falls into this class when $\xi = 0.5$.
- Class 2 are the functions with Smooth Multimodal Maps. An equation falls under this class when $0 < \xi < 0.5$.
- Class 3 are the piecewise continuous maps. An equation falls under this class when $1 \leq \xi \leq 1.5$

The derivation of the coefficient isolates smooth geometries D = 0 from non-smooth geometries featuring sharp corners or boundaries $D \geq 1$ by utilising the absolute signum function, $|sgn(D)|$ which returns the sign of D it can either have the value 0 or 1. When D = 0, the value of $|sgn(D)|$ becomes zero, which causes the left-hand term $\frac{1}{K+1}$. For a single smooth maximum K = 1, this evaluates to exactly 0.5, causing the value of ξ to fall under class 1. Higher-order stationary points $K \geq 2$ compress the output to $0 < \xi < 0.5$ intervals, which causes the equations to fall under class 2. Under the Class 3 structural constraints where K = 0, this active expression simplifies to $1.0 + \frac{0.5}{D}$, which mathematically causes the coefficient to stay within the explicit boundary of $1.0 < \xi < 1.5$ regardless of the corner density. This unified formulation encapsulates all structural map variations into a single, continuous mathematical

expression, ensuring deterministic class assignment based entirely on the intrinsic parameters of differentiability and extrema count. While the framework can be broken down into individual checks for D and K, the unified coefficient acts as a single continuous algorithmic function. This allows for a deterministic, automated classification of any discrete-time dynamical system into clear boundaries without requiring separate, branching conditional logic. For example in a scenario where a hybrid equation like the absolute-value sine map, $r|sin(x)|$, contains a mixture of shapes, featuring two sharp corners alongside one rounded peak, meaning D is 2 and K is 1. Instead of running multiple separate tests, the single coefficient formula handles both features simultaneously to output a value of 1.16, instantly grouping it into Class 3. Similarly, in a different scenario where a multi-segmented piecewise tent map, $r(1 - 2\left|x - \frac{1}{2}\right| + \frac{1}{10}sin(3\pi x))$, creates a complex curve with four sharp corners and three smooth valleys, meaning D is 4 and K is 3. The exact same formula automatically processes this combination of sharp and smooth attributes to generate a definitive value of 1.07, immediately classifying the system as a Class 3 behavior type without requiring manual decision trees.

3 Results and Discussion

The bifurcation diagrams for class 1 equations demonstrate the universal behaviours of smooth, single-peaked unimodal maps, in which each diagram begins with a stable fixed point and then splits into a stable two-cycle orbit at a precise parameter value that initiates the classic period-doubling route to chaos. This pitchfork branching process is repeated over smaller intervals, and the geometric ratio of successive interval widths always approaches the universal Feigenbaum constant of 4.669. It displays dense regions of chaotic attractors bounded by the maximum values of the functions after passing through the chaotic threshold. These chaotic zones contain a dense set of distinct vertical bands representing stable periodic windows, during which the system temporarily leaves chaos to follow a predictable cyclic path before returning to its original chaotic state.

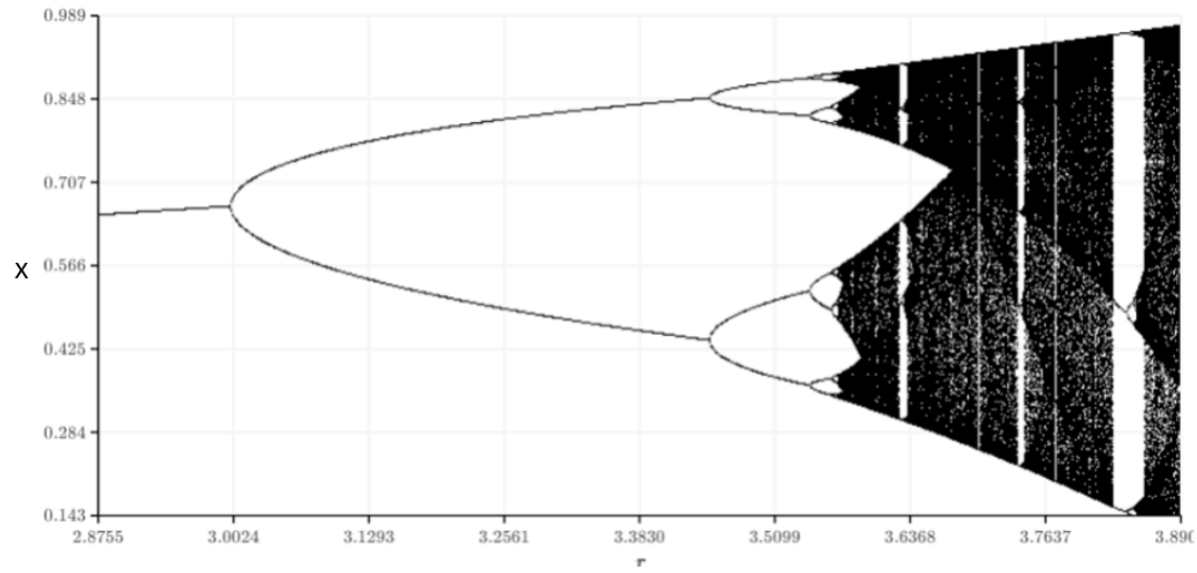


Figure 2: The bifurcation diagram for class 1 equation: rx(1-x)

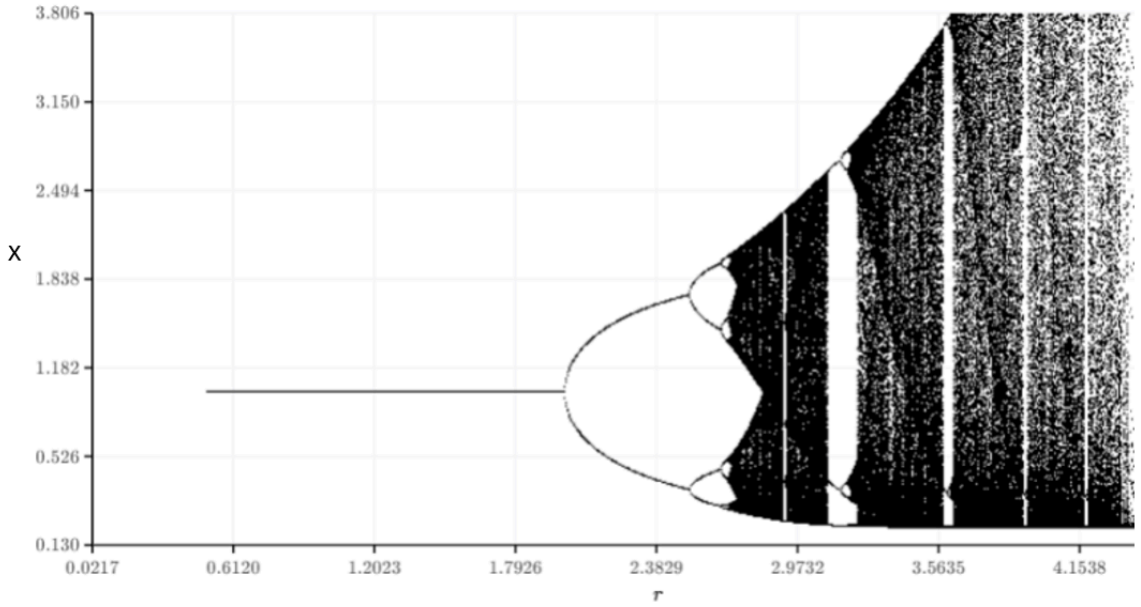


Figure 3: The bifurcation diagram for class 1 equation: $xe^{(r(1-x))}$

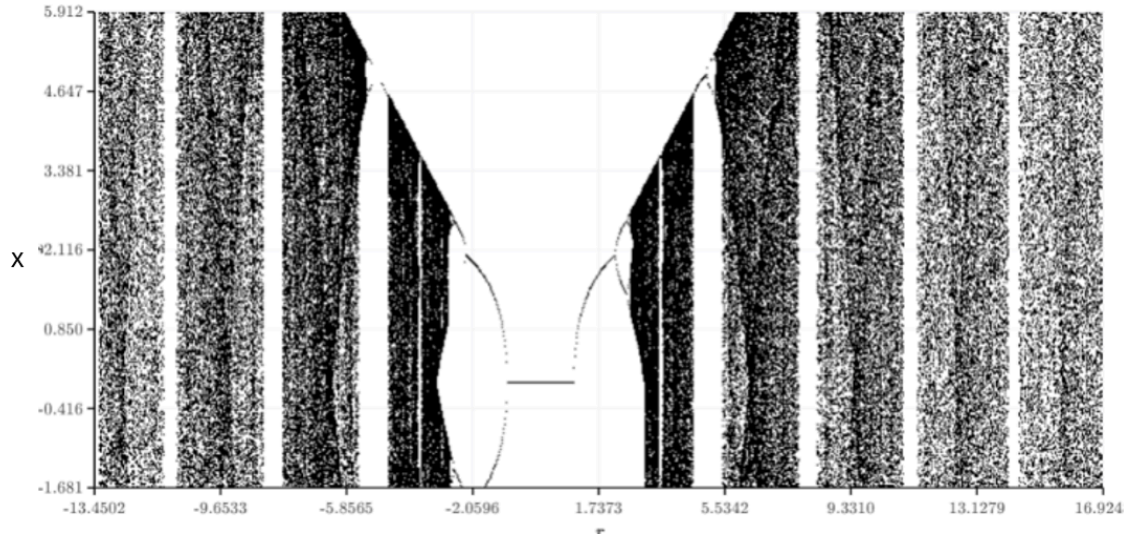


Figure 4: The bifurcation diagram for class 1 equation: rsin(x)

Equation	**Stable Equilibrium Range**	**First Period Doubling Bifurcation**	**Chaotic Threshold**	**Notable Periodic Window**
$rx(1-x)$	< 3.0	3.0024	3.56	3.83
$xe^{r(1-x)}$	≤ 2.0	2.0	2.69	3.1
$rsin(x)$	$[-1.7373, 1.7373]$	2.26	2.718	5.5342

Table 1: Quantitative bifurcation metrics and critical transition thresholds for smooth unimodal Class 1 ($\xi = 0.50$) *systems across variations in the control parameter r.*

The bifurcation diagrams, which display the universal properties of smooth, single-peaked unimodal maps. As seen in the standard logistic map trajectory ($rx(1-x)$)(Figure 2, Table 1), the system maintains a stable equilibrium below 3.0 before the first period-doubling bifurcation occurs at 3.0024. The post-threshold chaotic regime, which initiates at 3.56, is characterised by dense geometric attractor bands interspersed with distinct vertical stripes representing stable periodic windows, including a

prominent wide period-3 window visible near 3.83. Figure 3 is another example of the exponential map ($xe^{r(1-x)}$), which maintains a steady-state value for $r \leq 2.0$ until an initial pitchfork split at 2.0. The subsequent cascade of nested bifurcations scales at a geometric rate governed by the universal Feigenbaum constant of 4.669 before collapsing into a high-density attractor zone at 2.69, featuring a clear stable periodic window near 3.1. Figure 4 depicts the sinusoidal map ($rsin(x)$) . The system's intrinsic periodicity introduces a highly symmetric topology extending across both positive and negative parameter fields. A central baseline equilibrium is sustained within the interval $[-1.7373, 1.7373]$ until the absolute parameter threshold reaches 2.26, forcing the stable path to split into open parabolic branches that extend symmetrically. Following a rapid period-doubling cascade that scales via Feigenbaum metrics, full chaotic dynamics have reached the absolute parameter boundary of 2.718. Visually, this culminates in a dense distribution of bounded chaotic attractors segmented by repeating vertical columns of stable periodic windows, with a dominant periodic window cleanly isolated at 5.5342.

The bifurcation diagrams for class 2 equations transition to smooth, multimodal maps. These systems contain multiple internal turning points, which generate parallel period-doubling structures across the parameter range. As the control parameter increases, trajectories interact with multiple stationary points, causing global bifurcations and interior crises. These crises cause sudden, discontinuous vertical expansions of the chaotic attractor boundaries instead of continuous outer growth. Within the fully developed chaotic regions, the diagrams exhibited a dense distribution of state values with overlapping bands and distinct intersecting geometric structures. These regions feature unique wave-like shapes and criss-crossing tracking lines formed by trajectories after following the multiple local stationary points. Moreover, the width of these successive parameter intervals shrinks at a precise geometric rate matching the Feigenbaum constant of 4.669.

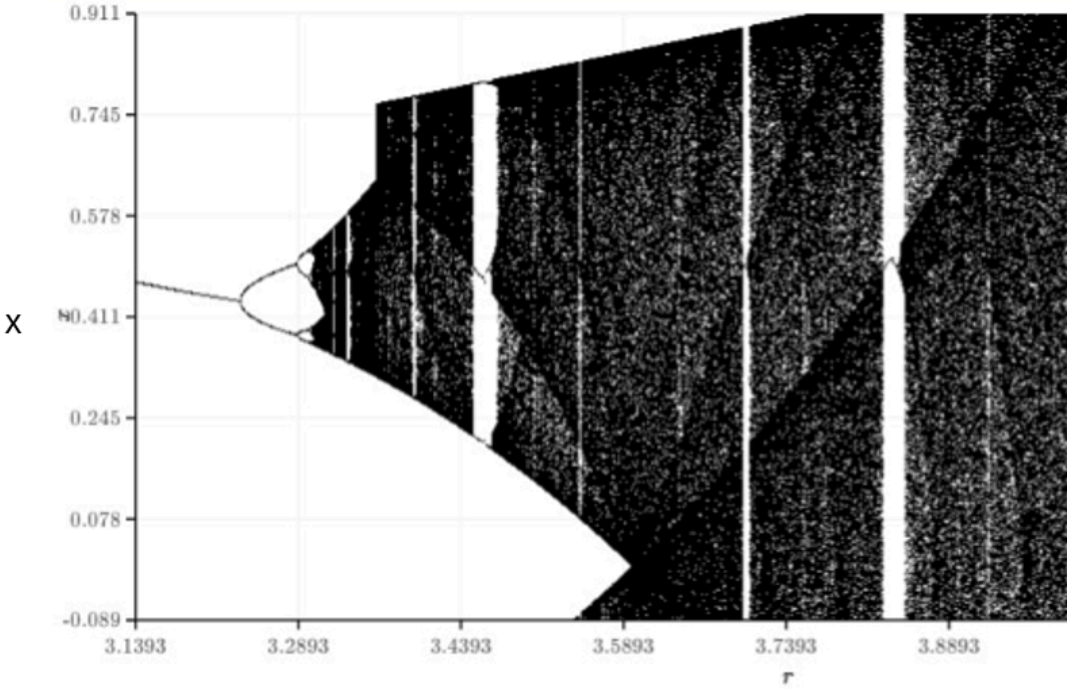


Figure 5: The bifurcation diagram for class 2 equation: $rx^3+(1-r)$

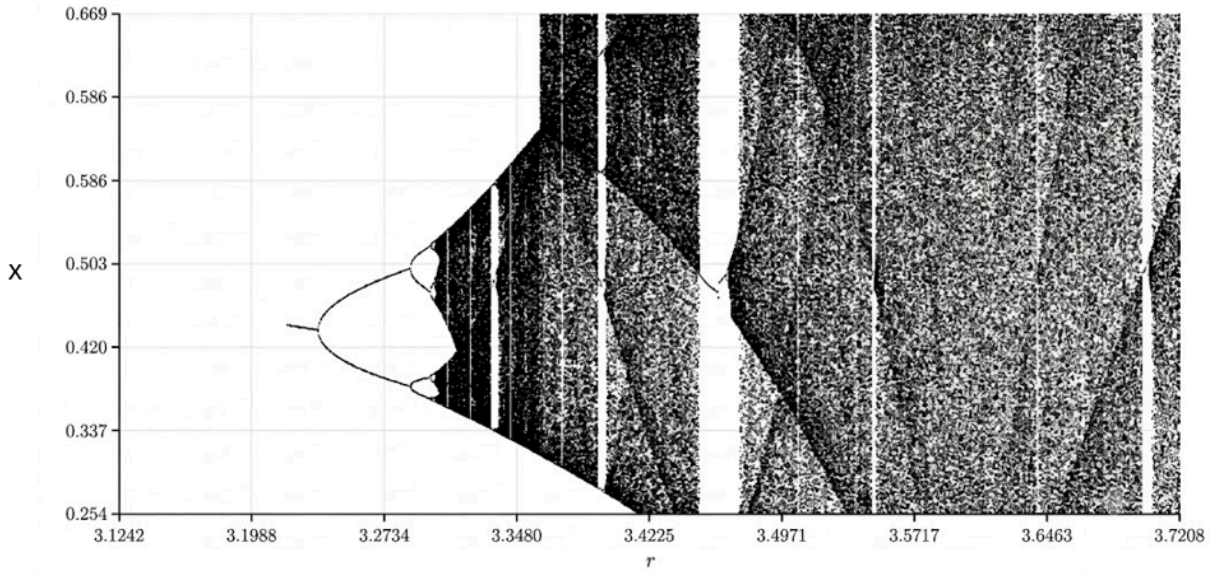


Figure 6: Bifurcation diagram for class 2 equation: $r(x^5-x^3+x)$

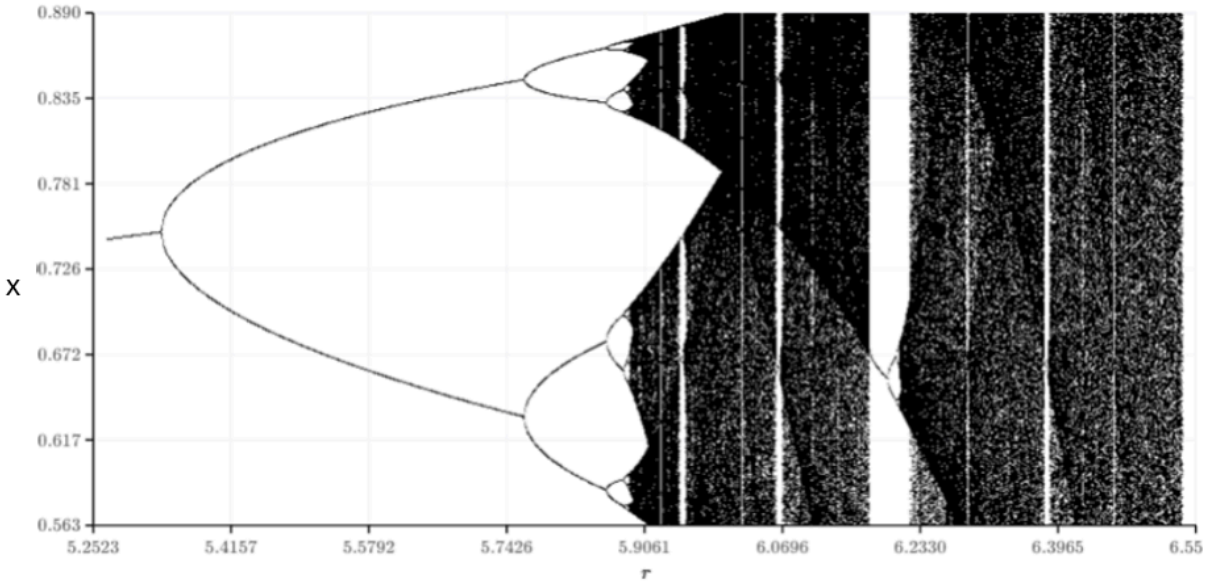


Figure 7: Bifurcation diagram for class 3 equation: rx^2 (1-x)

Equation	**First Period-Doubling Bifurcation**	**Chaotic Threshold**	**Notable Periodic Window**	**Attractor Bound Limits**
$rx^3 + (1 - r)$	3.0	3.33	3.83 (Period-3)	$[-0.089, 0.911]$
$r\left(x^5 - x^3 + x\right)$	3.23	3.31	3.46	$[0.254, 0.740]$
$rx^2(1 - x)$	5.34	5.90	6.15	$[0.563, 0.890]$

Table 2: Quantitative bifurcation metrics and critical transition thresholds for smooth multimodal Class 2 ($0 < \xi < 0.5$*) systems across variations of the control parameter r.*

This is clearly shown by the bifurcation topologies for Class 2 systems ($0 < \xi < 0.5$) that exhibit complex, higher-order structures characteristic of smooth multimodal maps containing multiple

local extrema. In the cubic map ($rx^3 + (1 - r)$), [Figure 5, Table 2], the system initiates with a stable two-cycle orbit that undergoes a cascade of nested bifurcations that scale at a geometric rate matching the Feigenbaum constant of 4.669 before collapsing into a full chaotic state at 3.33. Following this threshold, a structural transition stabilizes a dense chaotic zone strictly bounded between state values of 0.089 and 0.911. This displays distinct triangular and wave-like geometries near 3.59 alongside clear vertical periodic windows, most notably a wide period-3 window near 3.83. This behaviour shifts significantly in the quintic map ($r\left(x^5 - x^3 + x\right)$), [Figure 6, Table 2], where an initial single fixed point undergoes its first period-doubling event at 3.23 and enters chaos at 3.31. At 3.34, the system undergoes a sudden, discontinuous vertical expansion that expands its attractor boundaries between 0.254 and 0.740, weaving intersecting geometric "criss-cross" patterns through the dense chaos and revealing a prominent stable window at 3.46. Similarly, the alternate cubic map ($rx^2(1 - x)$)[Figure 7, Table 2] begins with a single stable path that splits into a two-cycle orbit at 5.34 and transitions to complete chaos at 5.90. The resulting chaotic domain is confined within state limits of 0.563 and 0.890 and is characterised by highly defined wave-like trajectories interspersed with stark vertical white columns of stable periodic windows, with the most dominant window neatly isolated at 6.15.

The general properties of Class 3 bifurcation diagrams define the structural transitions specific to non-smooth, piecewise linear maps. These systems lack smooth, differentiable maxima, eliminating the traditional period-doubling route to chaos. Instead, the diagrams demonstrate a direct transition from a single stable fixed point to a fully developed chaotic attractor at a critical parameter threshold. This sudden transition appears geometrically as a solid, expanding wedge of chaotic state values with a uniform point density. Since these systems bypass the infinite cascade of pitchfork bifurcations, they do not follow the geometric scaling of the Feigenbaum constant of 4.669. The chaotic wedge is bounded by sharp linear edges determined by the slopes of the piecewise functions, and it frequently features distinct lens-shaped or eye-shaped white gaps near the initial transition point, representing unvisited regions of the state space before a global bifurcation drives trajectories to infinity.

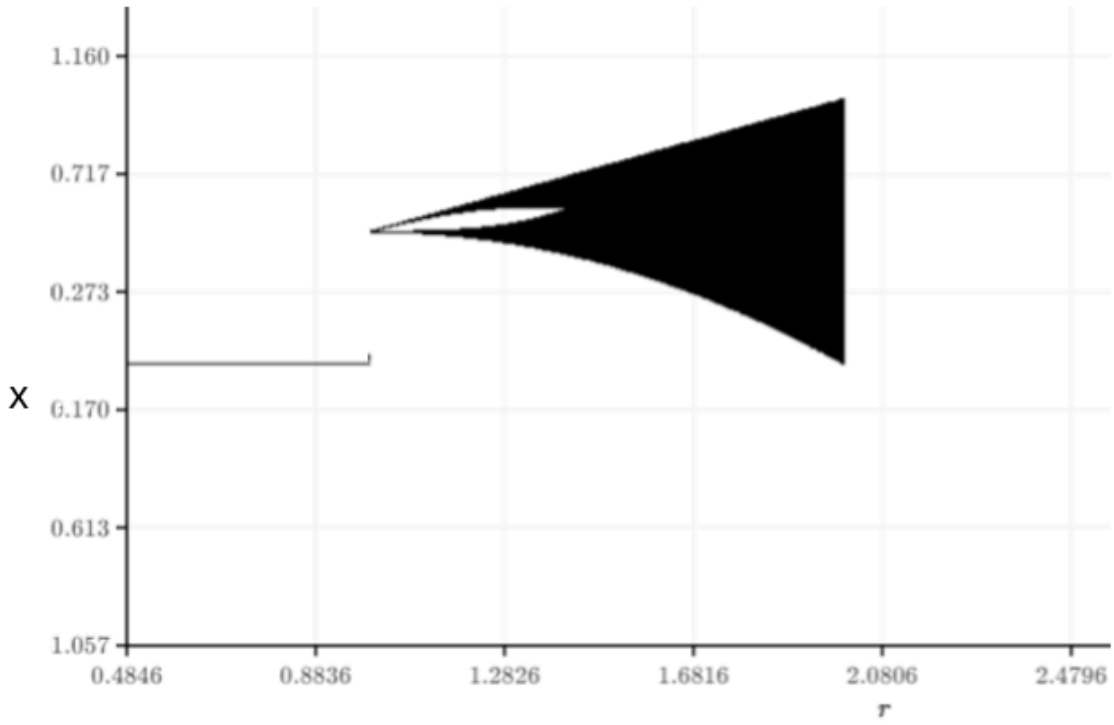


Figure 8: Bifurcation diagram for class 3 equation: $r(1 - 2|x - 0.5|)$

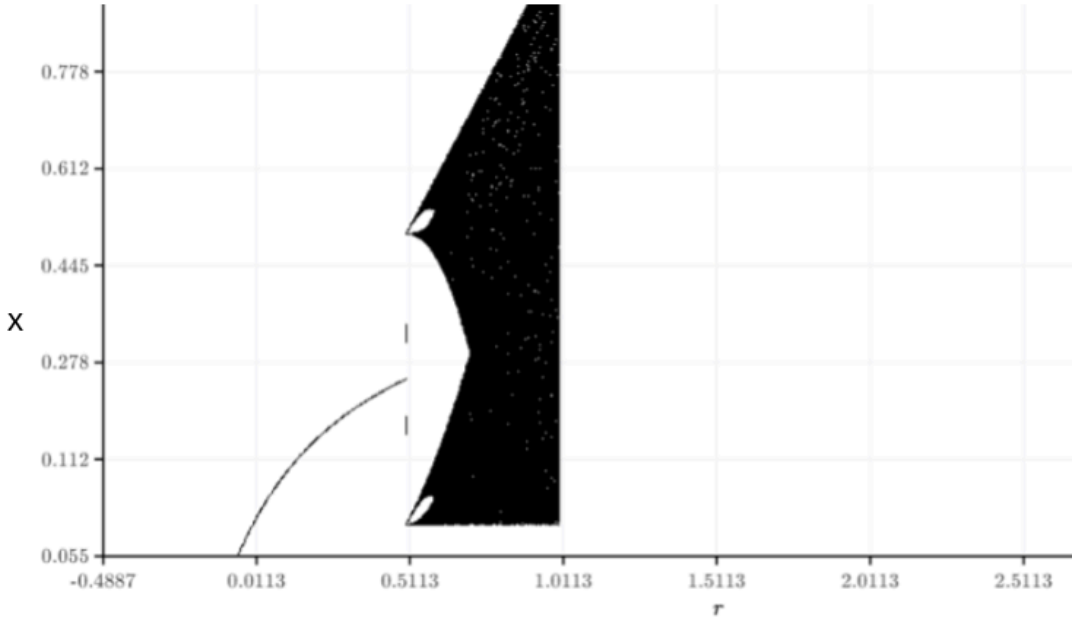


Figure 9: Bifurcation diagram for class 3 equation: $r|2x - 1|$

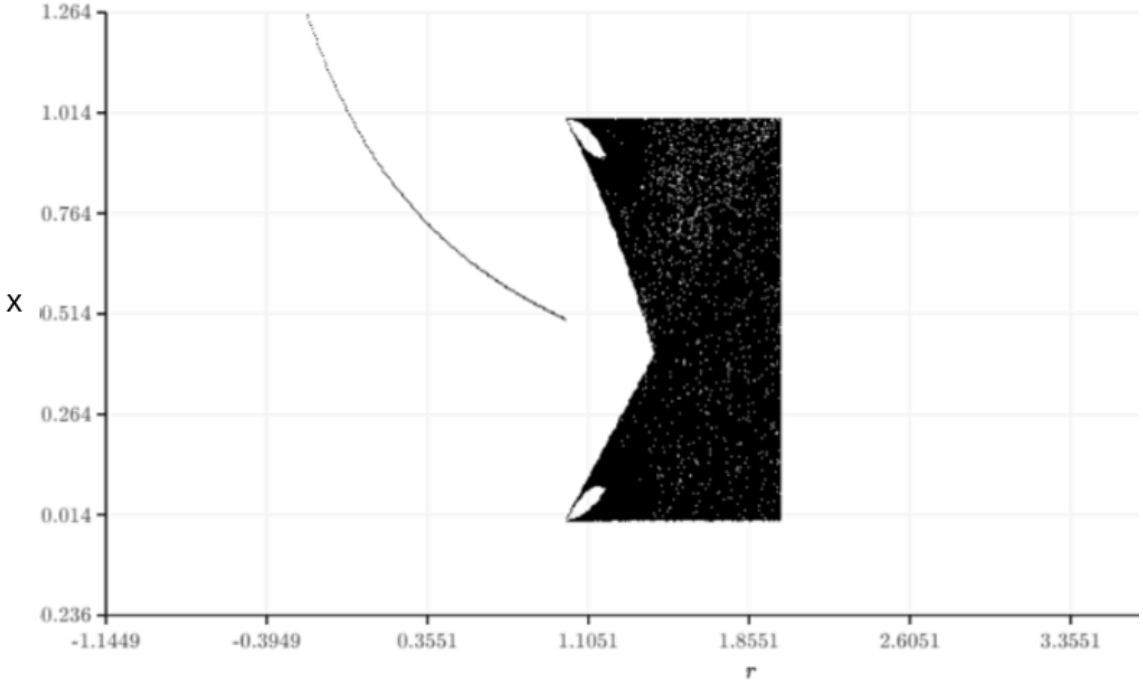


Figure 10: Bifurcation diagram for class 3 equation: $|1 - rx|$

Map Equation	**Transition Threshold (r)**	**Period-Doubling Cascade**	**Attractor Envelope (x)**	**Boundary Divergence (r)**
$r(1 - 2\|x - 0.5\|)$	1.0	Bypassed entirely	[-0.05, 1.00]	2.0
$r\|2x - 1\|$	0.5	Bypassed entirely	[0.00, 1.00]	1.0
$\|1 - rx\|$	1.0	Bypassed entirely	[0.00, 1.00]	2.0

Table 3: Quantitative bifurcation metrics and critical transition thresholds for equations with distinct geometries characteristic of non-smooth, piecewise linear maps with sharp corner boundaries for Class 3 ($1.0 < \xi < 1.5$*) systems across variations of the control parameter r.*

This is clearly shown by the bifurcation topologies for Class 3 systems $1.0 \leq \xi \leq 1.5$, which fundamentally depart from smooth configurations, displaying the distinct geometries characteristic of non-smooth, piecewise linear maps with sharp corner boundaries. In the linear tent map ($r(1 - 2|x - 0.5|)$), [Figure 8, Table 3], the system maintains a single stable fixed point near zero until the control parameter reaches 1.0, where it entirely bypasses the traditional period-doubling sequence to collapse directly into a full chaotic state. Following this sudden transition, a uniform chaotic zone is stabilized between state values of -0.05 and 1.00 until a global bifurcation drives trajectories to infinity at 2.0, leaving the dense black wedge completely uninterrupted by stable periodic windows except for a single, distinct lens-shaped white gap on the parameter interval [1.0, 1.4]. This behaviour shifts slightly in the sharp, V-shaped absolute value map ($r|2x - 1|$) [Figure 9, Table 3], where an initial stable fixed-point path terminates abruptly at 0.5, bypassing the standard cascade to plunge instantly into a fully developed chaotic attractor bounded between 0.00 and 1.00. The system operates with uniform density until a boundary crisis forces global divergence at 1.0, weaving two small, symmetrical eye-shaped phase-space voids near the initial 0.5 transition threshold. Similarly, the alternate absolute value map ($|1 - rx|$), [Figure 10, Table 3] begins with a gradually declining curvilinear fixed-point trace that drops out discontinuously at 1.0. The system skips the regular pitchfork path to generate a solid, homogeneous chaotic wedge confined within state limits of 0.00 and 1.00 before terminating at a global bifurcation limit of 2.0, with its uniform point distribution cleanly punctuated only by twin symmetric lens-shaped gaps opening near the initial parameter boundary.

It is clear that the different classes of equations can be identified by observing the unique structural and geometric properties of their bifurcation diagrams. The current experimental method exhibits clear structural and physical constraints that affect its accuracy across different map structures. The classification coefficient struggles to correctly identify systems with highly asymmetric turning points or maps possessing mixed continuity profiles, such as those that blend smooth quadratic curves with non-smooth sharp stationary points in separate state-space regions. Moreover, the coefficient range also introduces narrow periodic windows and coexisting attractors. Since the system's long-term behaviour depends entirely on its initial conditions, the system's final behaviour depends entirely on where it starts, causing its classification to shift sharply between chaotic and stable labels. We faced limitations due to insufficient computing power, which caps the simulation depth at 1500 iterations, a

constraint that prevents the elimination of initial transient states. This limit forces the system to sample unconverted trajectories, introducing noise and creating artificial points in unvisited regions, such as lens-shaped voids or periodic windows. Furthermore, the choice of the parameter step size directly limits the accuracy of the analysis. If the steps are too large, the system misses narrow stable windows and precise bifurcations. As a result, it miscalculates critical boundaries and underestimates how fast the system is changing. Consequently, these structural limitations and numerical constraints prevent this current framework from capturing or classifying the higher-order properties of $\xi > 1,5$ systems, indicating that further algorithmic classification and deeper analytical metrics are required.

The results of the experiment successfully fulfil the aim, which is to build a mathematical framework for classifying equations based on bifurcation diagrams and comparing the properties of these bifurcation diagrams across non-linear iterative equations. The data proves that the classification coefficient quantifies visual properties of bifurcation diagrams by linking algebraic form to dynamical regimes.

4 Conclusion

This study developed a mathematical framework for comparing chaos across nonlinear iterative equations. The structural classification coefficient ξ successfully quantified the visual properties of bifurcation diagrams by linking algebraic form to dynamical regimes. For Class 1 unimodal maps, the system tracked continuous pitchfork branches for the logistic map and sine map, which followed period-doubling cascades governed by the Feigenbaum constant of 4.669. For Class 2 multimodal maps, the data captured wave-like structures and sudden vertical boundaries caused by interior crises. For Class 3 piecewise linear maps, the system bypassed the infinite cascade completely and terminated the stable trace abruptly, and then expanded into a solid chaotic wedge containing an unvisited lens-shaped spot. Future research will extend this classification framework to analyse higher-order systems where the coefficient exceeds 1.5. Mathematicians should segregate this extended range into distinct sub-classes to map the specific properties of non-linear cusps. Additionally, researchers should isolate structural factors that influence the shapes of bifurcation diagrams to incorporate new parameters into the formula. They should also apply this metric to multi-dimensional systems to determine if the algebraic link to bifurcation geometry remains valid across higher-dimensional phase spaces.

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